CICC

IEEE Custom Integrated Circuits Conference

17.2: A 28-nm Real-Time Reinforcement Learning Processor for Mapless Autonomous Navigation with Unified Actor-Critic Network and Inference-on-Request Scheduling

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Outline

- Background and Challenges
- Proposed Processor
  - Feature 1: Unified Actor-Critic Network with Feature Map Caching
  - Feature 2: Inference-on-Request (IoR) Scheduling
  - Feature 3: Adaptive Dataflow Mapping
  - Feature 4: Column-wise Zero-Skipping Categorical Projection
- Measurement Results and Comparison
- Conclusion

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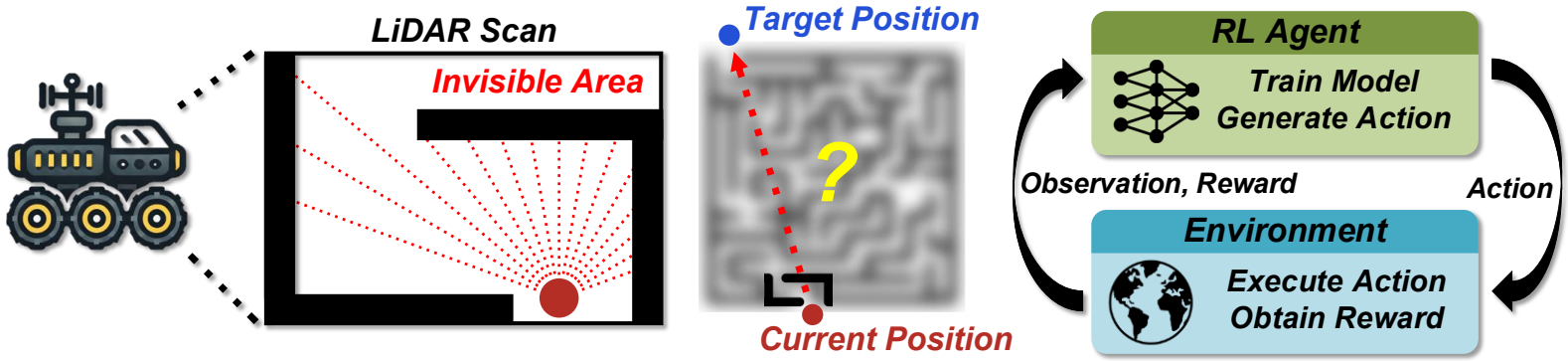
Mapless Autonomous Navigation

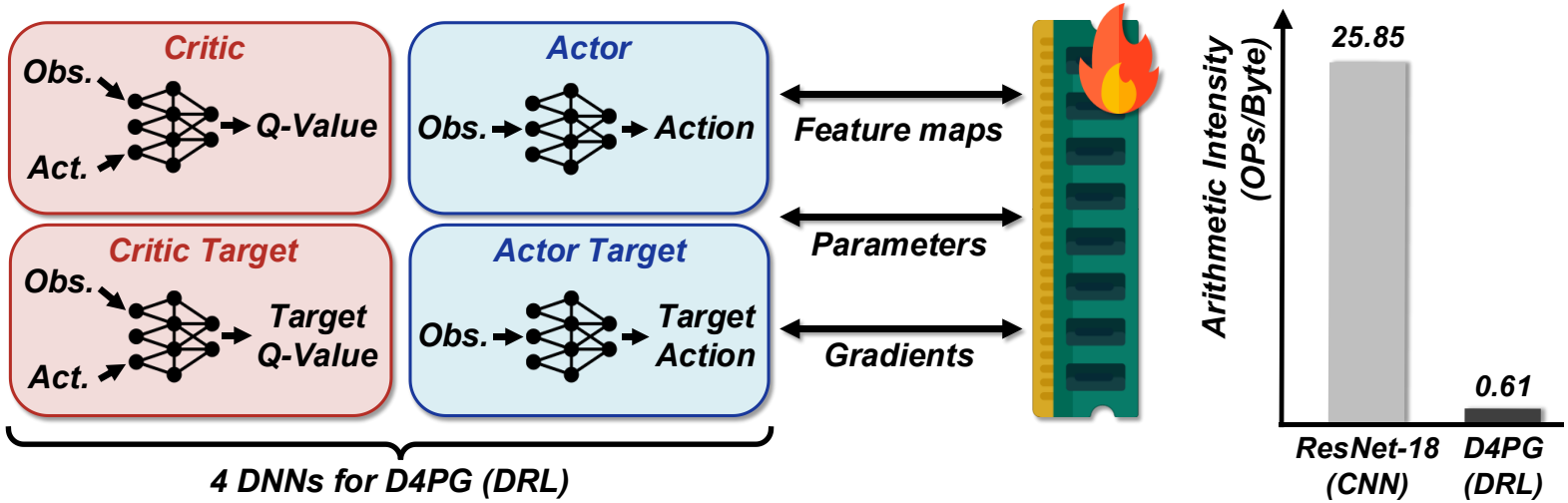
- Mapless navigation relies solely on local observations and target position information without the access to global map data
- Deep Reinforcement Learning (DRL) is a promising solution for mapless autonomous navigation



Challenges of DRL for Real-Time Navigation

1) Extensive External Memory Access

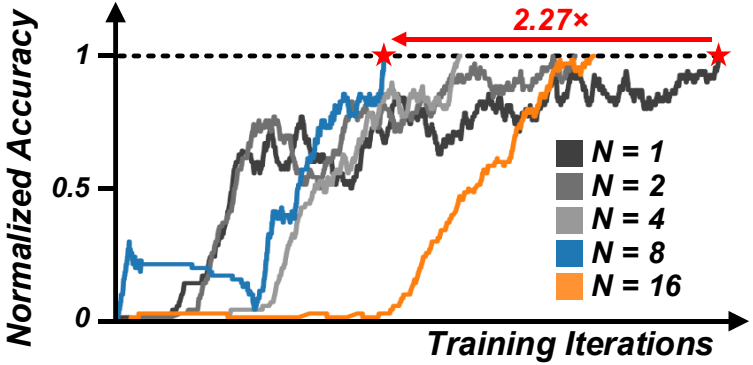
- **DRL typically relies on multiple Deep Neural Networks (DNNs)**
- **DNNs for DRL are mostly composed of fully connected layers**
  - Low arithmetic intensity, frequent external memory access

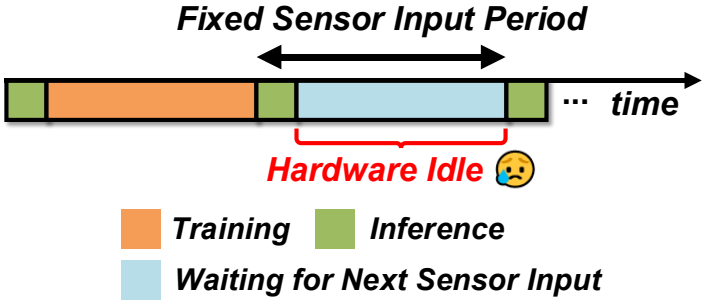


Challenges of DRL for Real-Time Navigation

2) Low Hardware Utilization in Sparse Training

- **Sparse training demonstrates significantly higher performance**
  - Training is conducted only after a set number of inferences (less frequently)
- **Naïve scheduling causes inevitable low hardware utilization**
  - Processor should perform training and inference simultaneously
  - Inference needs to be processed in sync with fixed sampling frequency of robot's sensor



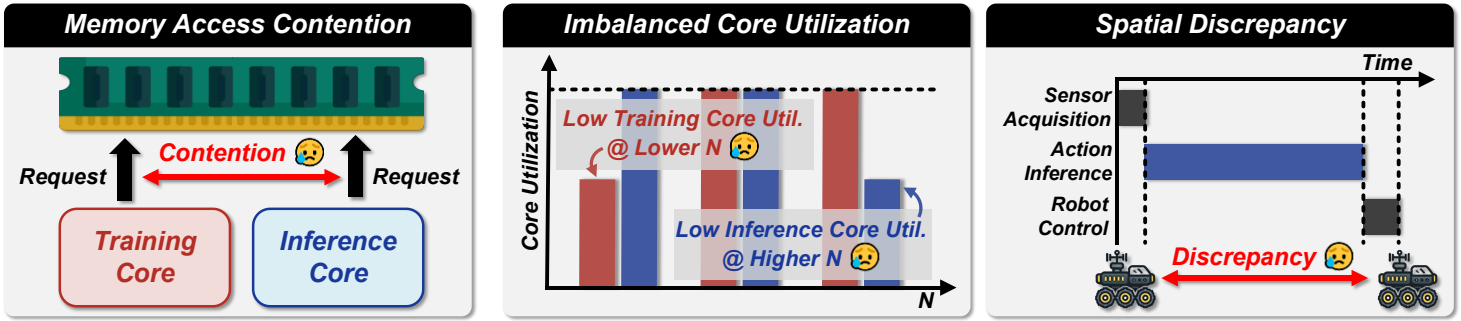


N: Inference-to-Training Ratio

Challenges of DRL for Real-Time Navigation

3) Limitations of Dedicated Training/Inference Cores

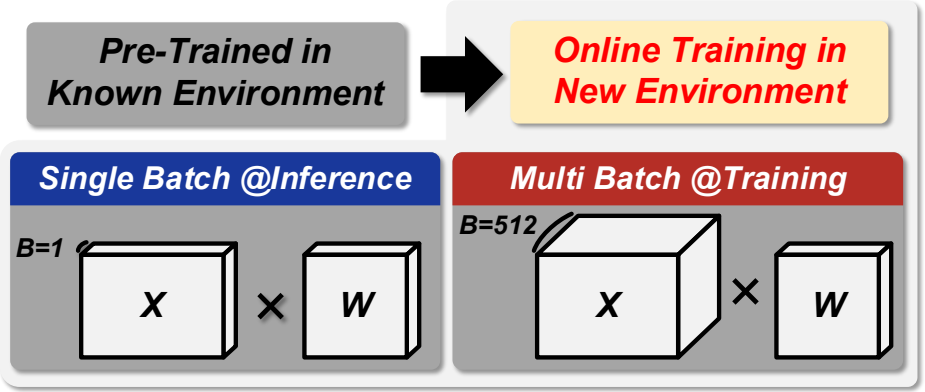
- **Simultaneous external memory access results in contention**
  - One core stalls while waiting for external data access due to bus contention
- **Limited flexibility at varying inference-to-training ratios (*N*)**
  - Full utilization of both cores is not guaranteed
- **Achieving full inference core utilization increases inference latency**
  - Prolonged action generation latency causes spatial discrepancy

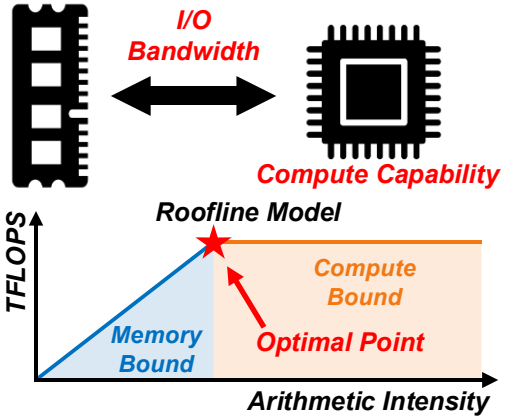


Challenges of DRL for Real-Time Navigation

4) Various Workload Characteristics

- **Training and inference have different workload characteristics**
  - Processor needs to efficiently support both single/multi-batch processing
- **I/O bandwidth and compute capability should be balanced**
  - All resources should be fully utilized during both training and inference





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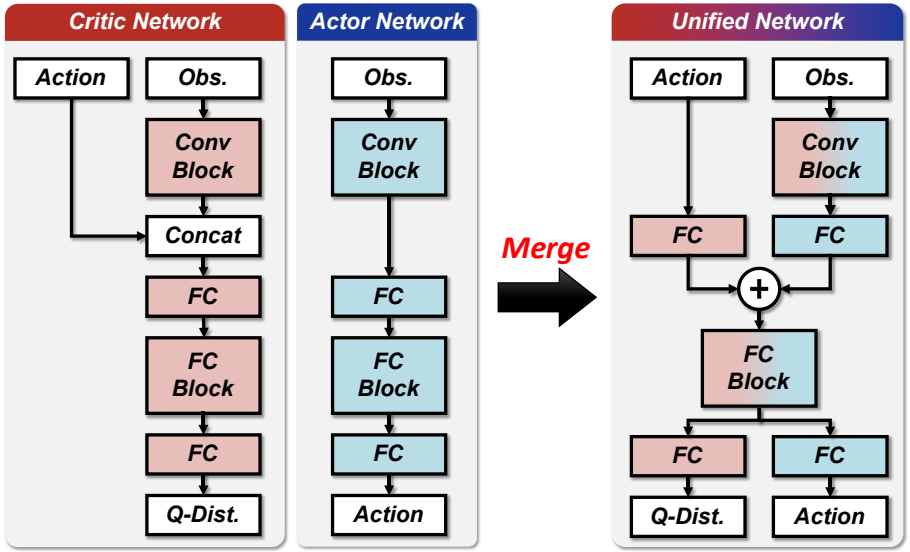
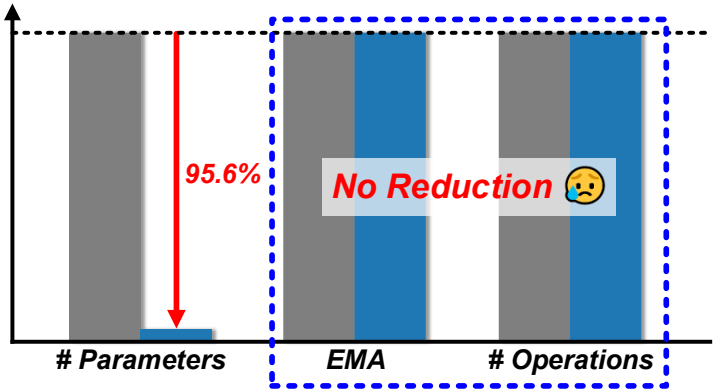
Unified Actor-Critic Network Architecture

Feature 1

- **Actor-Critic based algorithms use 2 kinds of neural networks**
- **Merge two distinct neural networks to exploit structural similarity**
  - But this only reduces parameters, excluding External Memory Access (EMA) and operations

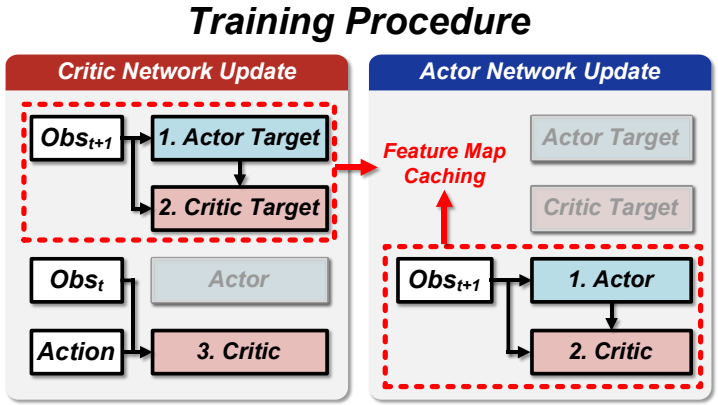


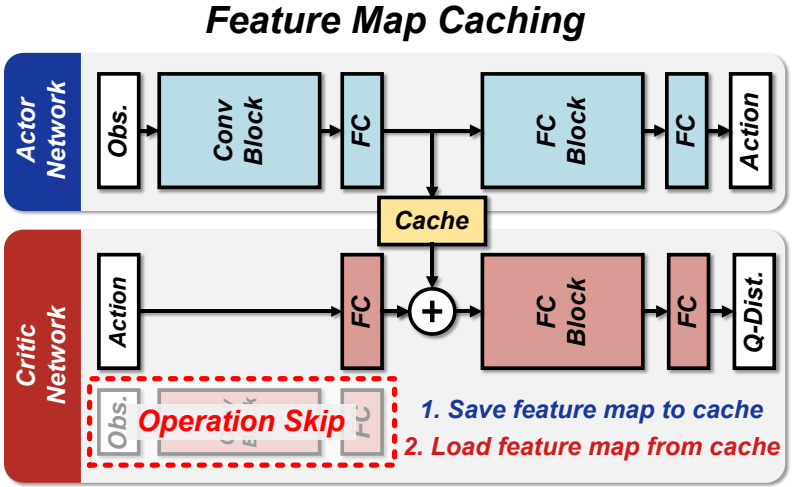


Feature Map Caching

Feature 1

- **Critic and Actor network is executed sequentially during the training**
  - Both networks take identical observations as an input
- **Caching the intermediate feature map from the actor network**
  - Reuse it in the critic network, avoiding duplicate computations

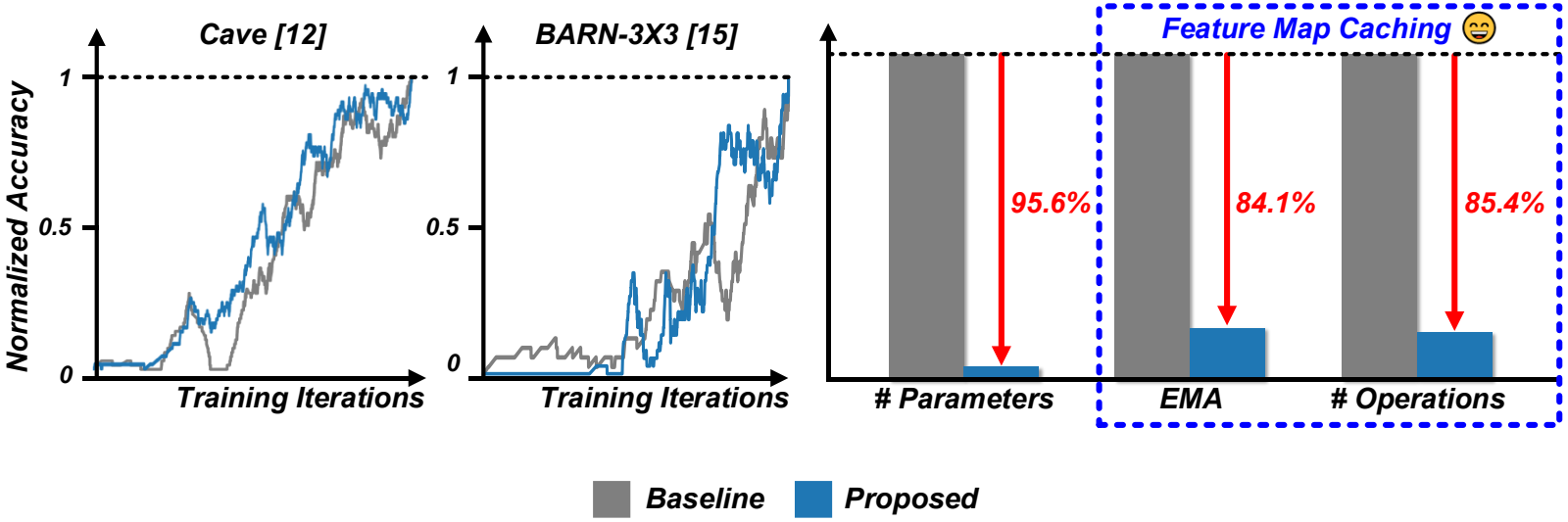




Performance Comparison

Feature 1

- Additional reduction of *EMA and operations by feature map caching
- Unified actor-critic network shows no performance degradation



*EMA: External Memory Access

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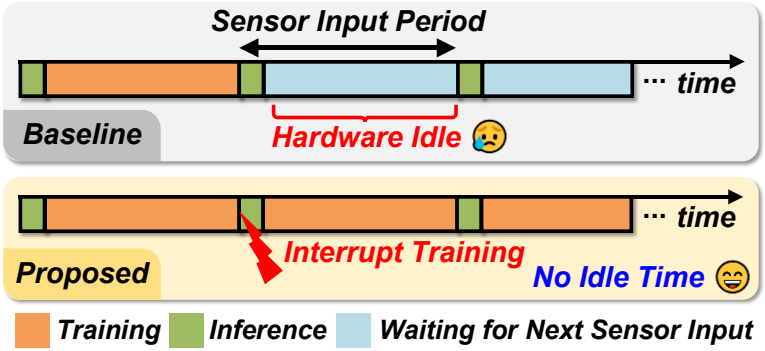
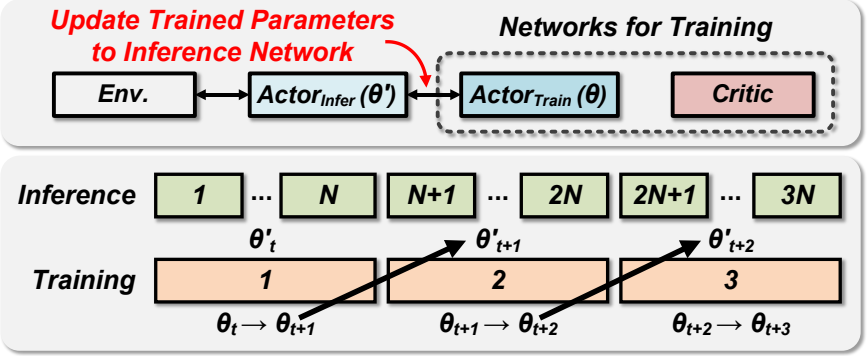
Inference-on-Request (IoR) Scheduling

Feature 2

- **Inference request interrupts the background training process**
  - Inference is prioritized and initiated by sensor input from the robot
- **Updated training parameters are copied to the inference network**
  - It avoids parameter coherence issues between training and inference



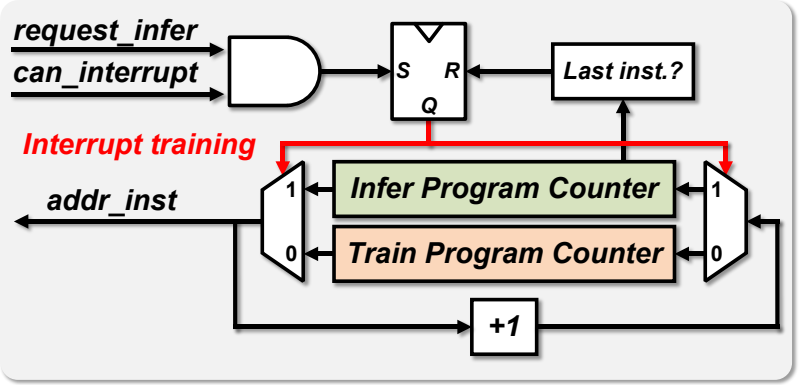


Instruction Management Unit (IMU) Operation

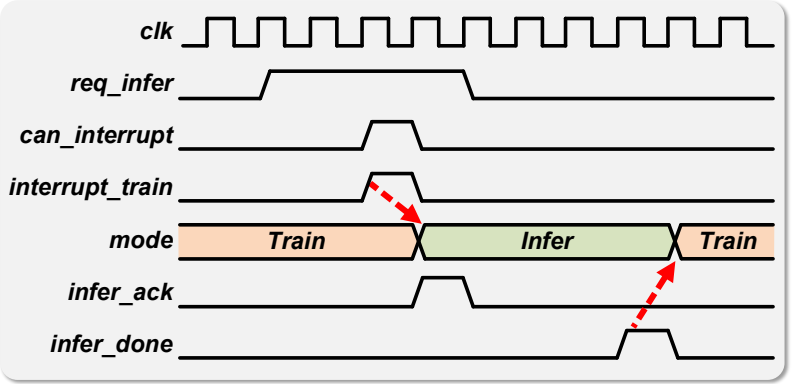
Feature 2

- **Instruction Management Unit (IMU) manages IoR scheduling**
  - It determines whether training instruction can be interrupted or not
- **IMU interrupts training once on-chip data reuse is no longer possible**
  - It avoids redundant external memory accesses

Instruction Management Unit (IMU)



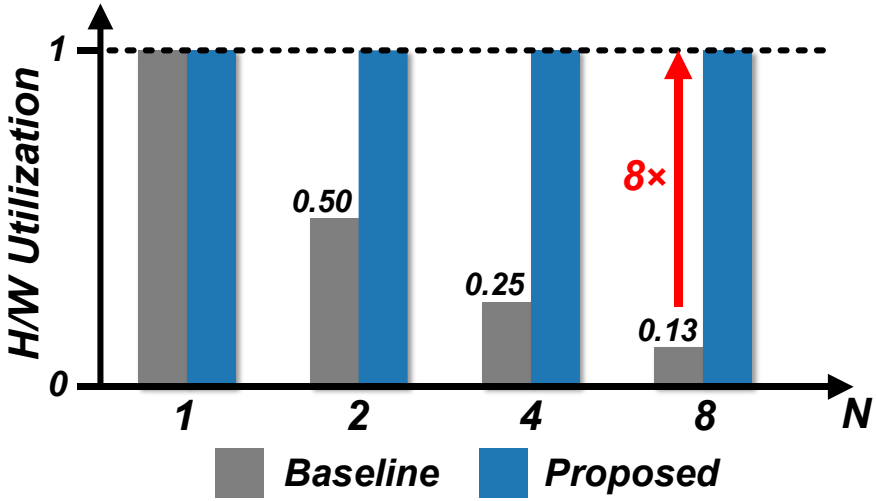
IoR Timing Diagram

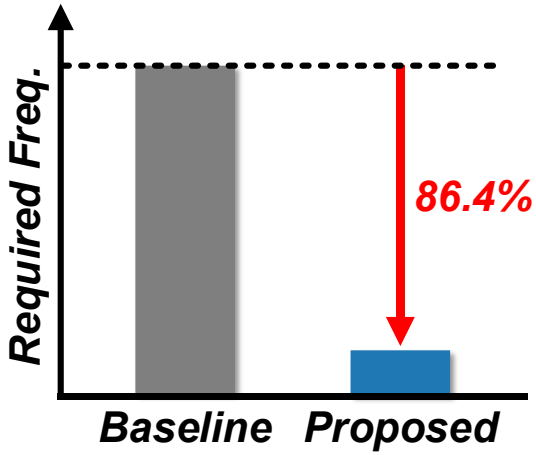


Performance of IoR Scheduling

Feature 2

- **Full hardware utilization regardless of the inference-to-training ratio (*N*)**
- **Required frequency for real-time operation reduced by 86.4%**
  - Worst inference latency < 10% of the sensor input period





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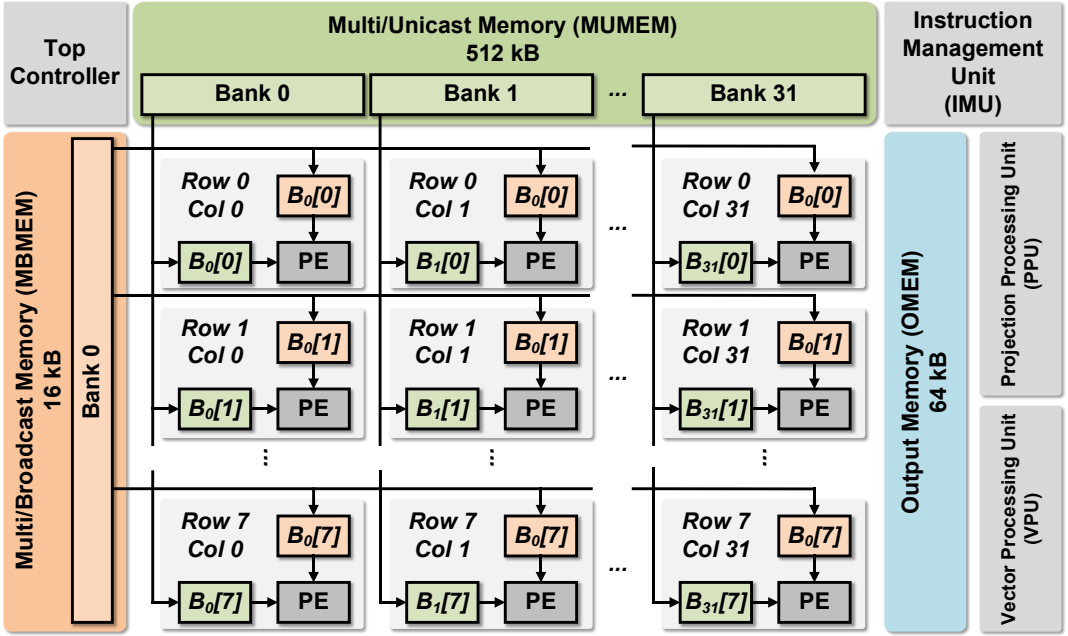
Overall Architecture

Feature 3

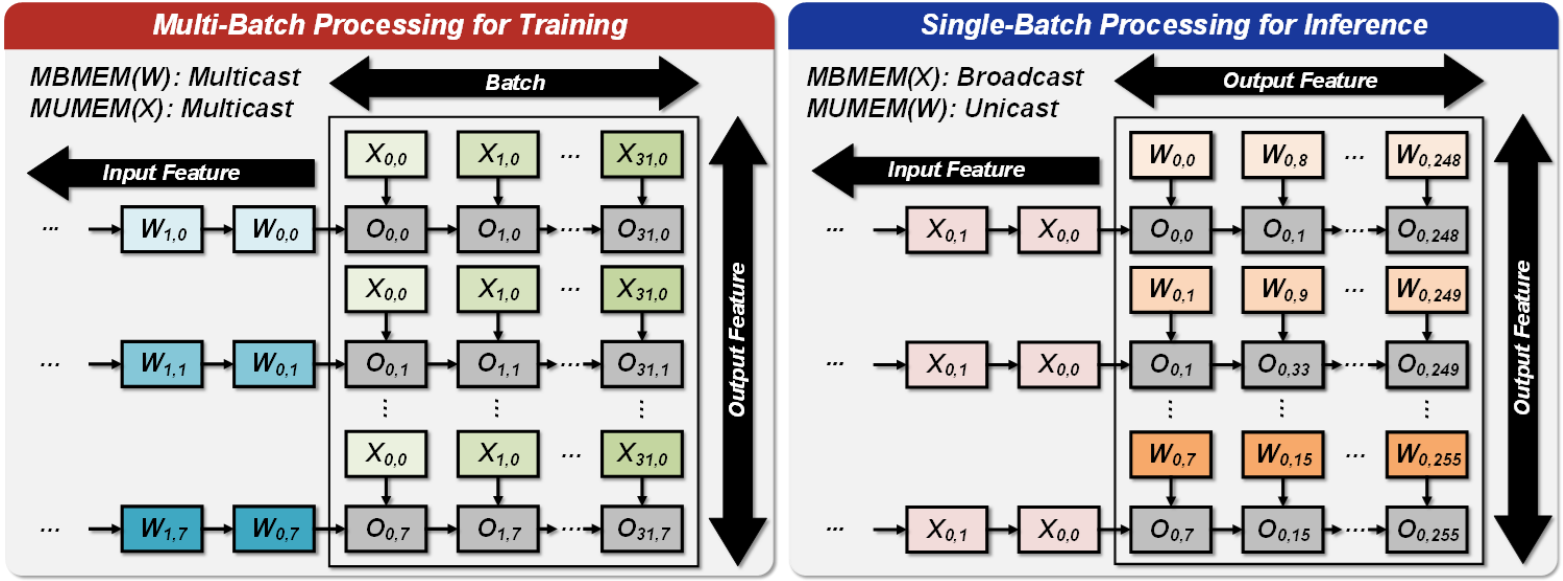
- **8×32 PE Array**
  - BF16 Multiply + FP32 Acc.
- **592-kB SRAM**
  - 512-kB Multi/Unicast Mem.
  - 16-kB Multi/Broadcast Mem.
  - 64-kB Output Mem.
- **Vector Processing Unit**
- **Projection Processing Unit**



Adaptive Dataflow Mapping

Feature 3

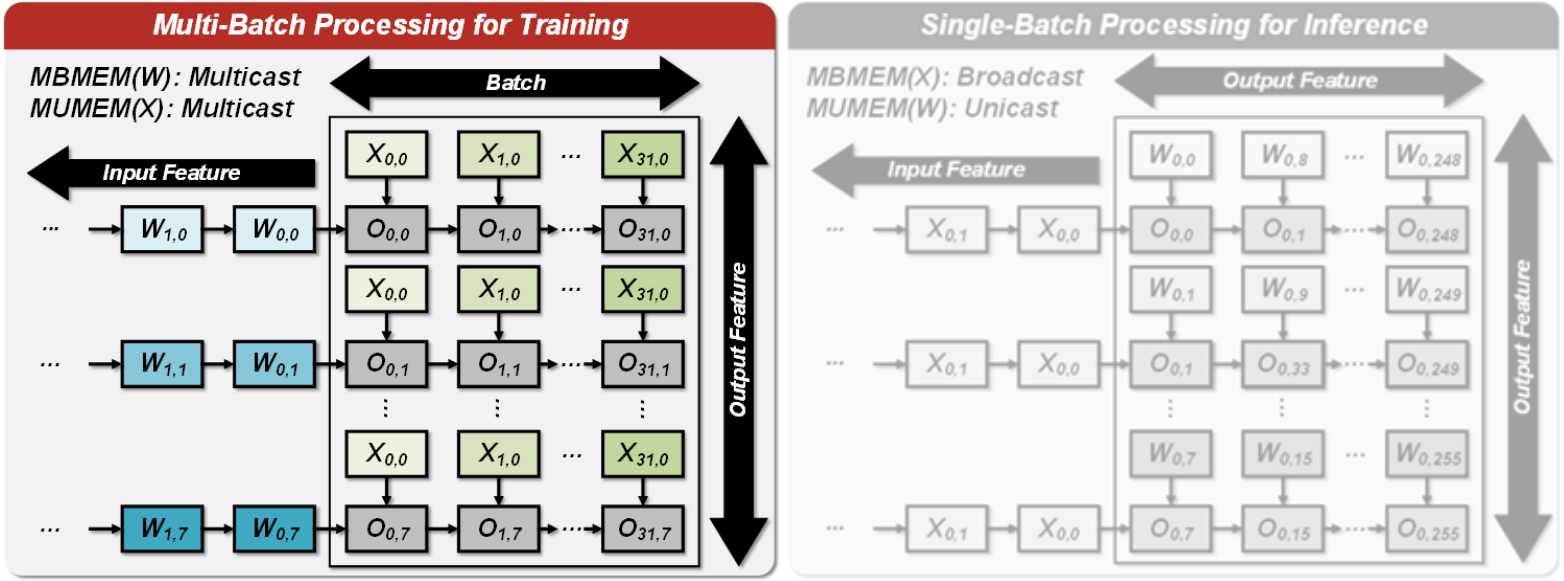
- **MBMEM and MUMEM distribute data differently depending on batch size**
  - MBMEM: Multi/Broadcast Memory, MUMEM: Multi/Unicast Memory



Multi-Batch Processing for Training

Feature 3

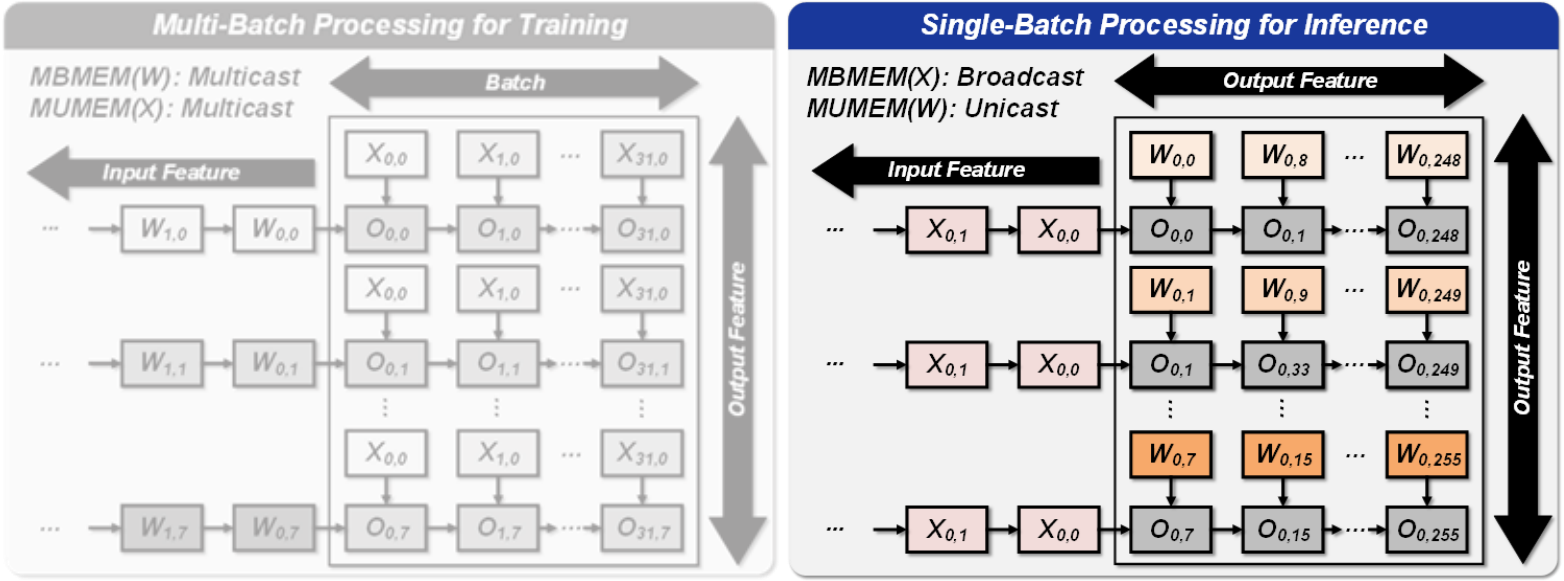
- MBMEM: Multicasts weights to PE array
- MUMEM: Multicasts input features to PE array



Single-Batch Processing for Inference

Feature 3

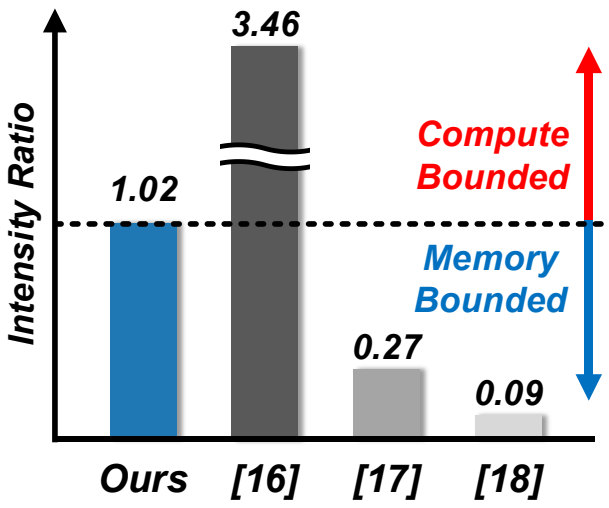
- MBMEM: Broadcasts input features to PE array
- MUMEM: Unicasts weights to PE array

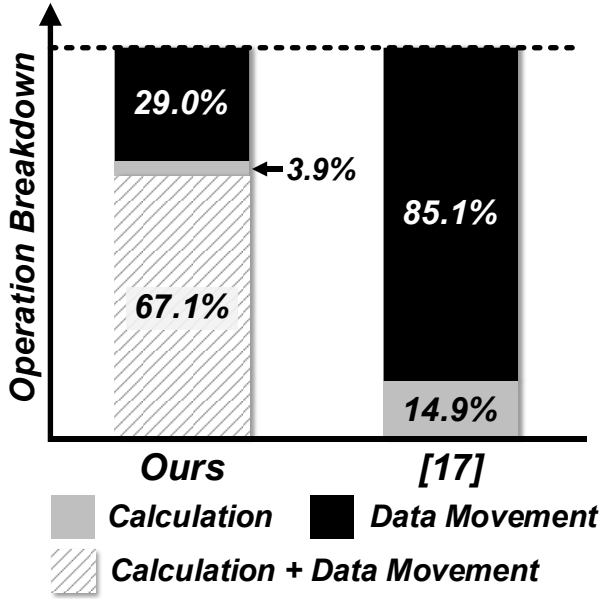


Arithmetic Intensity Matched Architecture

Feature 3

- **Arithmetic intensities of processor and algorithm are matched**
  - High PE utilization of 71.0% achieved including I/O latency during the entire training scenario





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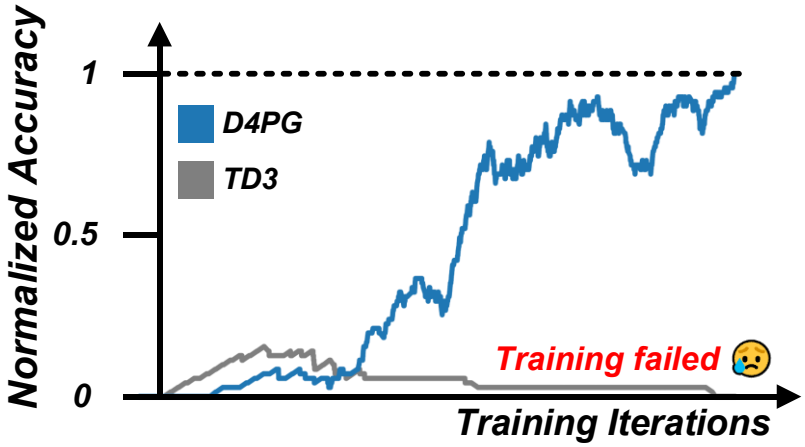
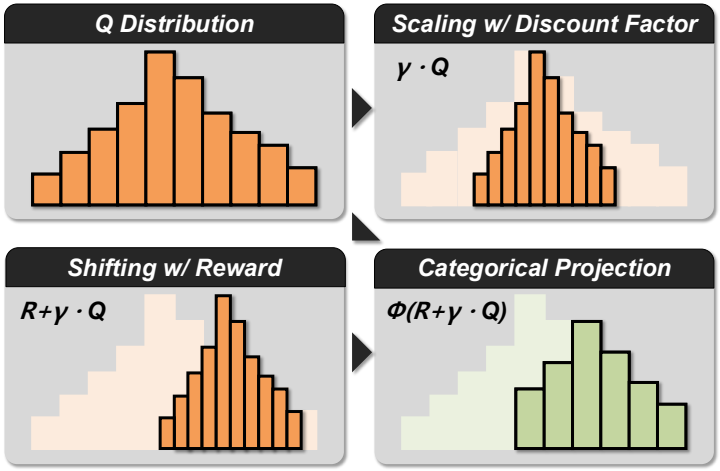
Target Distribution Generation for Distributional RL

Feature 4

- **Distributional RL provides superior performance to traditional RLs**
  - Utilizes Q-distribution instead of scalar Q-value
- **Q-value distribution is transformed with discount factor (γ) and reward (R)**
  - Redistribution is needed for the loss calculation

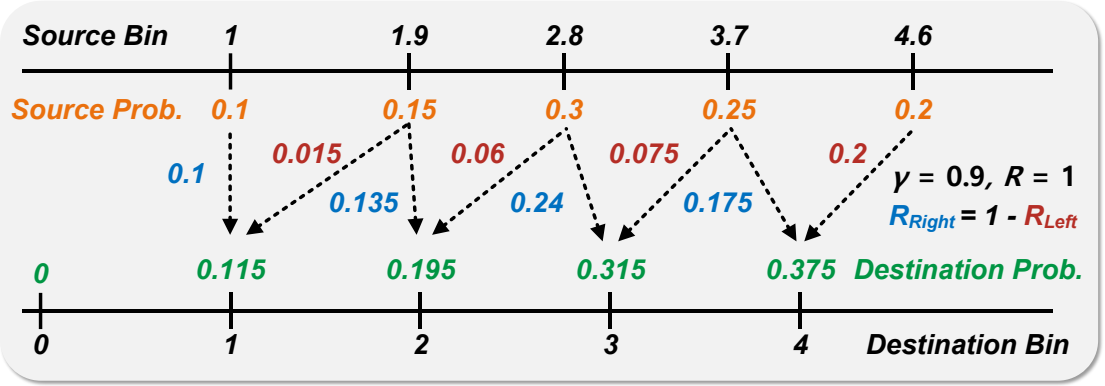


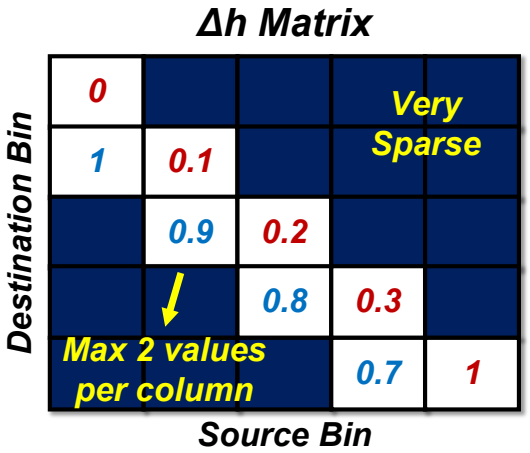


Categorical Projection

Feature 4

- **Redistribute the probabilities across bins**
  - Account for affine-transformed probabilities
- **Δh matrix represents the probability transfer amounts**
  - It is very sparse (At most 2 values per column)

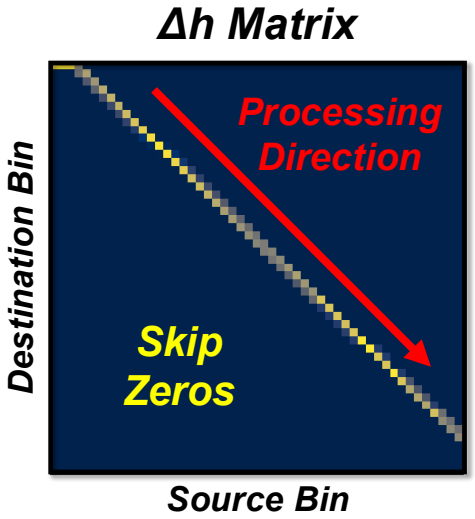


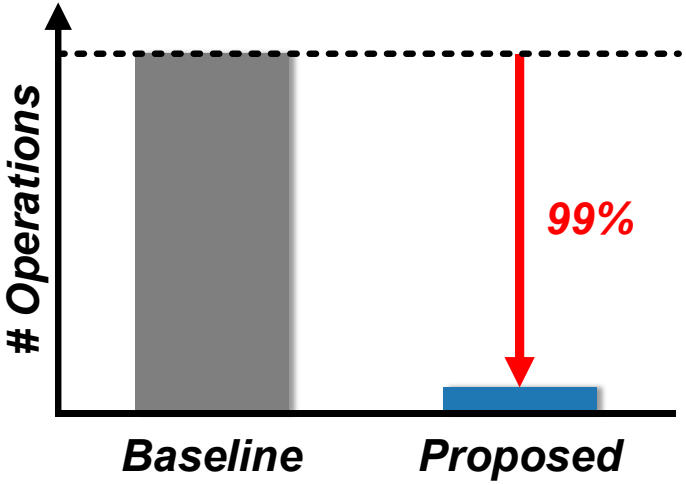


Column-Wise Zero-Skipping Categorical Projection

Feature 4

- **Process one column of Δh matrix per cycle**
  - Fetch and process only non-zero values in each column
- **Total projection operations reduced by 99% for 51 bins**





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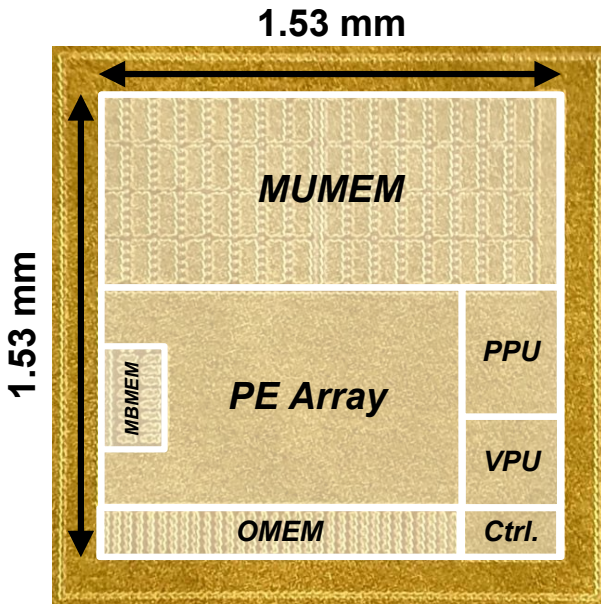
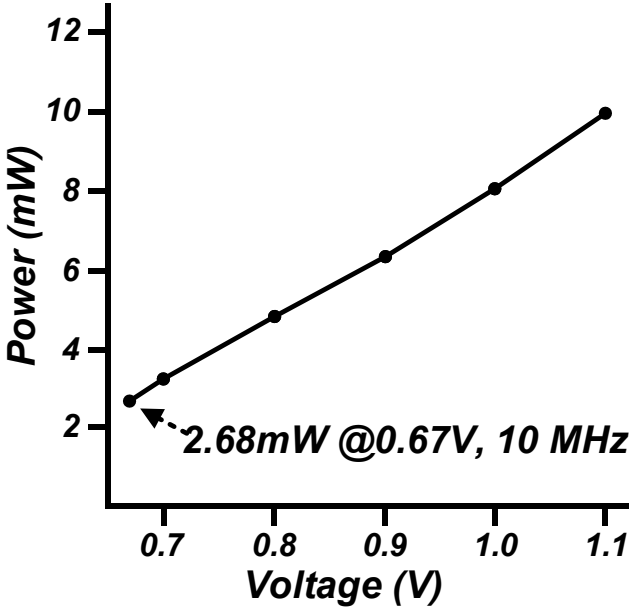
Die Photo and Measurement Result

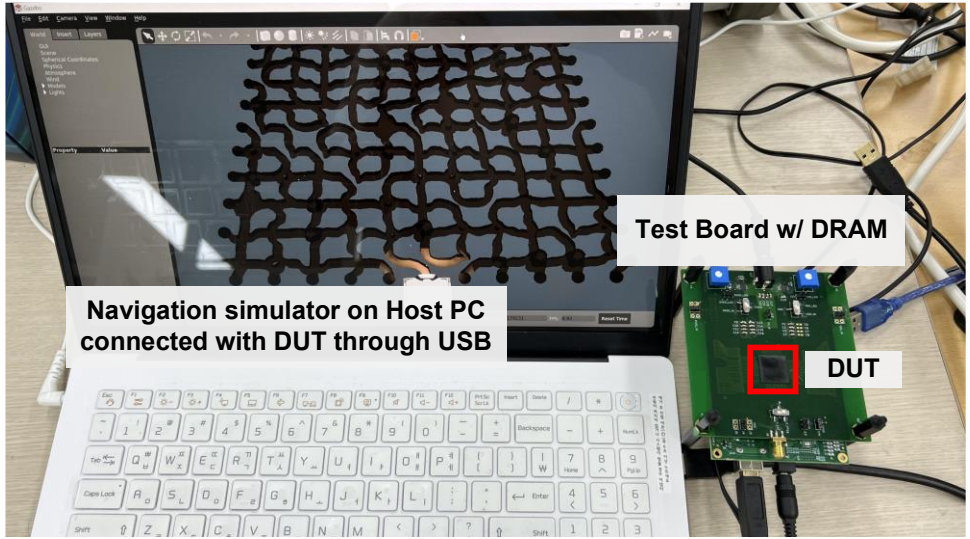
- Our processor only consumes 2.68mW with real-time operations

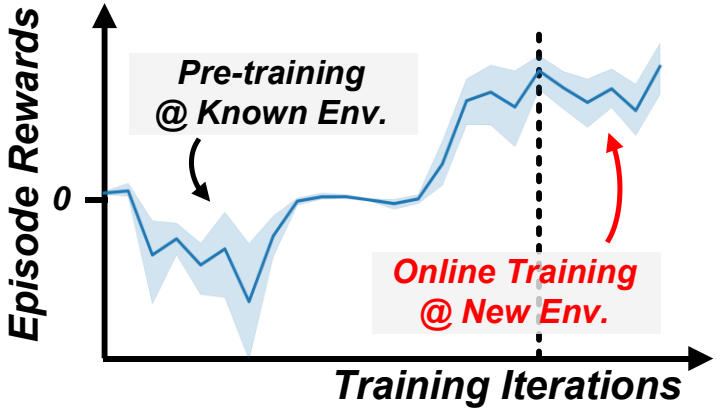




Simulation Setup and Result

- **LDO-based power regulation + FPGA-based clock generation**
  - Standalone operation without external clock generator and power supply is available





Comparisons with Prior Work

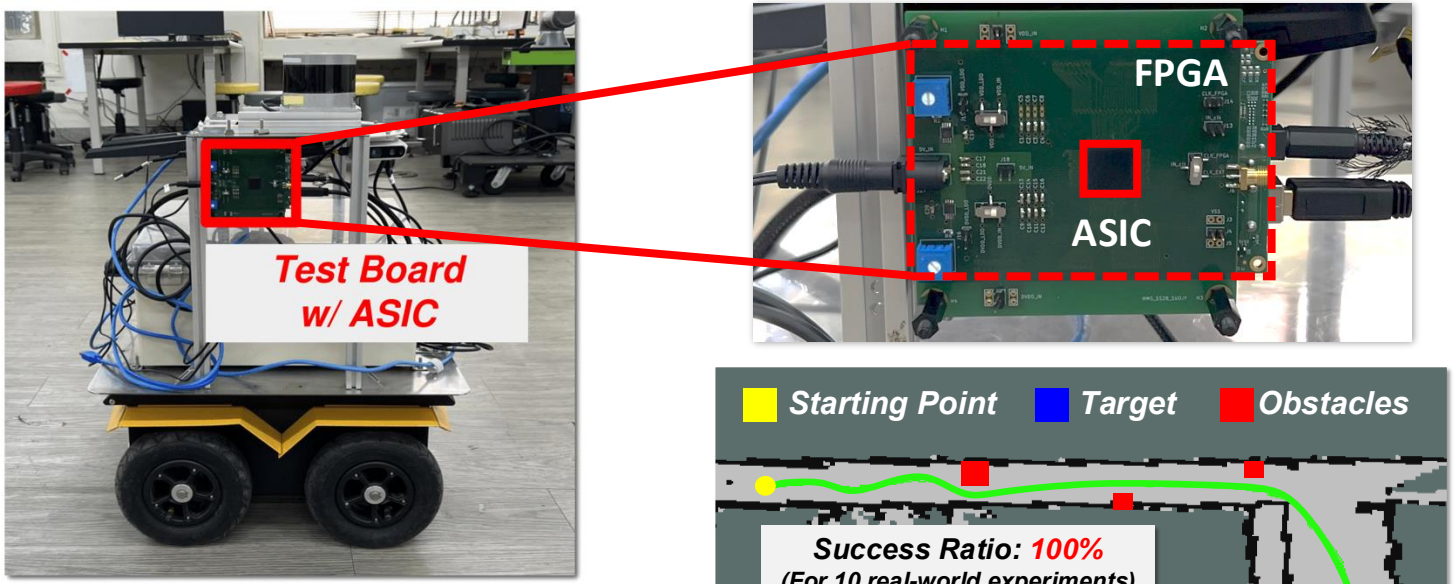
		This Work	ISSCC'19 [16]	VLSI'21 [17]	TCAS-II'24 [18]
Verified on Real Robot		**O**	X		
Support for Distributional RL		**O (D4PG)**	X		
Technology (nm)		**28**	65	28	28
Area (mm^2)		**2.34**	16[a]	28[a]	12.96[a]
Precision	**Weight**	**Bfloat16**	FXP4/8/16	Bfloat16	BFP12/16
	Activation	**Bfloat16**	Bfloat16	Bfloat16	BFP12/16
SRAM (kB)		**592**	448	1552	2594
EMA Reduction		**84.1%**	41.0%[b]	51.5%[c]	45.1%[c]
Arithmetic Intensity (OPs/Byte)	**Hardware**	**32.0**	127.5	404.0	1025.0
	Model	**32.5**	441.0[b]	108.6[c]	90.8[c]
Intensity Ratio		**1.02**	3.46	0.27	0.09
Processor Utilization for Sparse Training[d]		**100.00%[e]**	12.51%[b]	12.61%[c]	12.61%[c]
Voltage (V)		**0.67 - 1.1**	0.67 - 1.1	0.68 - 1.1	0.9
Power Consumption (mW)		**2.68 (10 MHz, 0.67 V)**	2.4 (10 MHz, 0.67 V)	6.7 (5 MHz, 0.68 V)	176.7 (200 MHz, 0.9 V)

** All results assuming 0% weight sparsity* [a] *Die Area* [b] *Actor-Critic, Hill Climbing, Batch 2048* [c] *TD3, MuJoCo HalfCheetah, Batch 256*

[d] $\frac{T+N}{N(T+1)}$, $T = \frac{\text{\# Operations for Training}}{\text{\# Operations for Inference}}$, $N = 8$ [e] *Batch 512*

Real-World Evaluation

- **Real-world evaluation using Jackal UGV at NYCU, Taiwan**



Evaluation Setup

Measured Trajectory

Real-World Evaluation Demo Video

- Real-world evaluation demo video link (YouTube)

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Conclusion

- **Reinforcement Learning Processor for Mapless Autonomous Navigation**
  - **Unified Actor-Critic Network Architecture with Feature Map Caching**
    - Parameters, EMA, and operations reduced by **95.6%**, **84.1%**, and **85.4%**
  - **Inference-on-Request (IoR) Scheduling**
    - **100%** hardware utilization across all inference-to-training ratios
    - Required operating frequency reduced by **86.4%**
  - **Adaptive Dataflow Mapping**
    - High PE utilization of **71.0%** including I/O latency during a training iteration
  - **Column-wise Zero-Skipping Categorical Projection**
    - Projection operations reduced by **99%**
- **Verified in real-world environment**
  - Proposed processor only consumes **2.68mW** during concurrent training and inference

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